
Reproducing a DDPG-Based Stock Trading Strategy

Yassine Meliani¹ George Kontorosis¹

Abstract

TBD

1. Background and Motivation

Recent work has modeled stock trading as a Markov decision process, allowing reinforcement learning methods to directly optimize sequential trading decisions rather than relying on separate return forecasting and portfolio optimization stages (Liu et al., 2018). In the target paper, Liu et al. formulate the trading state as current stock prices, portfolio holdings, and remaining cash balance, and train a Deep Deterministic Policy Gradient (DDPG) agent to maximize changes in portfolio value over time (Liu et al., 2018). They report that this approach outperforms both the Dow Jones Industrial Average (DJIA) and a classical min-variance portfolio baseline in terms of cumulative return and Sharpe ratio (Liu et al., 2018). However, financial reinforcement learning results are often sensitive to modeling assumptions, hyperparameters, and market regime changes, especially in non-stationary environments. We therefore propose not only to reproduce their DDPG framework, but also to test how robust the reported gains are under targeted ablations, a stronger modern variant of the algorithm, and more recent market data including the COVID period.

Methods We will first reproduce the experimental setting of Liu et al. as closely as possible, using the same universe of 30 Dow Jones stocks and the same historical daily-price split described in the paper: training on 2009–2014 data, validation on 2015 data, and trading/testing on 2016–2018 data (Liu et al., 2018). We will then extend the study by evaluating the same framework on more recent data through 2026, allowing us to test whether the learned policy remains effective during substantially different market conditions, including the COVID-19 crash and recovery period. The environment will follow the paper’s MDP formulation, where

each state contains stock prices, current holdings, and available balance, actions correspond to buy/sell/hold decisions across assets, and rewards are defined by changes in portfolio value (Liu et al., 2018). We will reuse the released repository for data handling and environment infrastructure, but re-code the core DDPG learning algorithm ourselves, including the actor-critic networks, replay buffer, target networks, and policy update mechanism following the original DDPG formulation (Lillicrap et al., 2015; AI4Finance Foundation, 2018). Experiments will be run on rented cloud hardware through Lambda Labs, using a single NVIDIA A100 GPU to support training and ablation studies efficiently. We will evaluate performance using the same benchmarks and metrics as the paper: namely DJIA, min-variance portfolio allocation, final portfolio value, annualized return, annualized volatility/risk, and Sharpe ratio (Liu et al., 2018). In addition to reproduction, we will run ablations on key design choices, including whether continued learning during the test period materially changes results, how sensitive performance is to exploration noise and target-update settings, and whether replacing DDPG with the more stable TD3 algorithm improves robustness and risk-adjusted returns (Fujimoto et al., 2018).

References

- AI4Finance Foundation. Deep reinforcement learning for stock trading: Ddpq algorithm. <https://github.com/AI4Finance-Foundation/Deep-Reinforcement-Learning-for-Stock-Trading-DDP> 2018. Accessed: 2026-03-19.
- Fujimoto, S., van Hoof, H., and Meger, D. Addressing function approximation error in actor-critic methods, 2018. URL <https://arxiv.org/abs/1802.09477>.
- Lillicrap, T. P., Hunt, J. J., Pritzel, A., Heess, N., Erez, T., Tassa, Y., Silver, D., and Wierstra, D. Continuous control with deep reinforcement learning, 2015. URL <https://arxiv.org/abs/1509.02971>.
- Liu, X.-Y., Xiong, Z., Zhong, S., Yang, H., and Walid, A. Practical deep reinforcement learning approach for stock trading, 2018. URL <https://arxiv.org/abs/1811.07522>.

¹Department of Computer Science, McGill University, Montreal, Canada. Correspondence to: Yassine Meliani <yassine.meliani@mail.mcgill.ca>, George Kontorosis <george.kontorosis@mail.mcgill.ca>, Eric Elmoznino <eric.elmoznino@mila.quebec>.